

## Curriculum Vitæ — Peter Haine

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Citizenship              *United States*

### Employment

Fall 2025–                **Assistant Professor**, University of Southern California  
2023–2025              **NSF Postdoctoral Fellow**, UC Berkeley  
                              *Sponsoring scientist*: David Nadler  
2022–2023              **Member**, Institute for Advanced Study  
                              *Sponsoring scientist*: Jacob Lurie  
2021–2022              **NSF Postdoctoral Fellow and UC President’s Postdoctoral Fellow**, UC Berkeley  
                              *Sponsoring scientist*: David Nadler

### Education

2021                      **Massachusetts Institute of Technology**, Cambridge, MA, *Ph.D. Mathematics*  
                              *Advisor*: Clark Barwick  
2016                      **Massachusetts Institute of Technology**, Cambridge, MA, *S.B. Mathematics*

### Research Interests

Homotopy theory, algebraic geometry, & related subjects

### Short-term Visits

Fall 2026                **Motivic Homotopy Theory: Connections and Applications**, SLMath, Berkeley, CA  
June 2025                **IMPAN Gdańsk**, Sopot, Poland  
March 2025              **Copenhagen Centre for Geometry and Topology**, Copenhagen, Denmark  
July 2024                **Institut de Recherche Mathématiques Avancées**, Strasbourg, France  
June 2024                **Copenhagen Centre for Geometry and Topology**, Copenhagen, Denmark  
June 2023                **Max-Planck-Institut für Mathematik**, Bonn, Germany  
June–July 2022          **Universität Regensburg**, Regensburg, Germany  
June 2022                **Higher Categories and Categorification, Part 2**, MSRI/Unidad Cuernavaca del Instituto de Matemáticas UNAM, Cuernavaca, México

Spring 2020                      Higher Categories and Categorification, MSRI, Berkeley, CA

### Selected Awards & Distinctions

|             |                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------|
| 2021–2025   | National Science Foundation Mathematical Sciences Postdoctoral Research Fellowship<br>NSF Award DMS-2102957         |
| 2021–2022   | President’s Postdoctoral Fellowship, University of California                                                       |
| 2021        | Mathematics Community Building Award, MIT                                                                           |
| 2020–2021   | George Lusztig PRIMES Mentorship, MIT                                                                               |
| Fall 2019   | AMS Graduate Student Travel Grant                                                                                   |
| August 2019 | Leibniz Graduate Student Grant, Mathematisches Forschungsinstitut Oberwolfach<br>Workshop on <i>Homotopy Theory</i> |
| 2016–2021   | National Science Foundation Graduate Research Fellowship                                                            |
| 2016–2019   | Dean of Science Fellow, MIT                                                                                         |

### Publications & Preprints

18. Q. Bai and P. J. Haine, *Localizing invariants of constructible sheaves*, Dec. 2025, arXiv:2512.12810.
17. P. J. Haine, T. Holzschuh, M. Lara, C. Mair, L. Martini, and S. Wolf, *The condensed homotopy type of a scheme*, Oct. 2025, With an appendix by B. Zavyalov. arXiv:2510.07443.
16. P. J. Haine, M. Porta, and J.-B. Teyssier, *Standard t-structures*, Apr. 2025, arXiv:2504.07473.
15. P. J. Haine, M. Ramzi, and J. Steinebrunner, *Fully faithful functors and pushouts of  $\infty$ -categories*, Mar. 2025, arXiv:2503.03916.
14. M. Carlson, P. J. Haine, and S. Wolf, *A generalization of Voevodsky’s reconstruction theorem of schemes from their étale topoi*, Selecta Math. (N.S.), vol. 31, no. 4, Paper No. 65, 37, 2025. DOI: 10.1007/s00029-025-01054-5, arXiv:2407.19920.
13. P. J. Haine, M. Porta, and J.-B. Teyssier, *Exodromy beyond conicality*, Jan. 2024, arXiv:2401.12825.
12. P. J. Haine and P. Pstrągowski, *Spectral weight filtrations*, Sep. 2023, arXiv:2309.15072.
11. P. J. Haine, T. Holzschuh, and S. Wolf, *Nonabelian basechange theorems and étale homotopy theory*, J. Topol., vol. 17, no. 4, Paper No. e70009, 45, 2024. DOI: 10.1112/topo.70009, arXiv:2304.00938.
10. P. J. Haine, *Profinite completions of products*, May 2024, To appear in *Algebraic and Geometric Topology*. arXiv:2406.00136.
9. P. J. Haine, T. Holzschuh, and S. Wolf, *The fundamental fiber sequence in étale homotopy theory*, Int. Math. Res. Not., no. 1, pp. 175–196, 2024. DOI: 10.1093/imrn/rnad018, arXiv:2209.03476.
8. A. Amabel, A. Debray, and P. J. Haine, Eds., *Differential cohomology: Categories, characteristic classes, and connections*, Jan. 2023, arXiv:2109.12250. To be published in *Cambridge Studies in Advanced Mathematics*.
7. P. J. Haine, *From nonabelian basechange to basechange with coefficients*, J. Pure Appl. Algebra, vol. 229, no. 7, Paper No. 107993, 30, 2025. DOI: 10.1016/j.jpaa.2025.107993, arXiv:2108.03545.
6. P. J. Haine, M. Porta, and J.-B. Teyssier, *The homotopy-invariance of constructible sheaves*, Homology Homotopy Appl., vol. 25, no. 2, pp. 97–128, 2023. DOI: 10.4310/hha.2023.v25.n2.a6, arXiv:2010.06473.
5. P. J. Haine, *On the homotopy theory of stratified spaces*, Ann. Sci. Éc. Norm. Supér. (4), vol. 57, no. 2, pp. 351–384, 2024, arXiv:1811.01119.

4. C. Barwick, S. Glasman, and P. J. Haine, *Exodromy*, Aug. 2020, arXiv:1807.03281. Includes the contents of the four papers arXiv:1812.11637, arXiv:1901.09414, arXiv:1904.01877, and arXiv:1904.09966.
3. S. Devalapurkar and P. J. Haine, *On the James and Hilton–Milnor Splittings, and the metastable EHP sequence*, Doc. Math., vol. 26, pp. 1423–1464, 2021, arXiv:1912.04130.
2. V. Guillemin and P. Haine, *Differential forms*. World Scientific Publishing Co. Pte. Ltd., Hackensack, NJ, 2019, pp. xvi+256, ISBN: 978-981-3272-77-4. DOI: 10.1142/11058.
1. S. Fomin, R. Shankar, P. Haine, and V. Chugunov, *Stability analysis of non-Newtonian rimming flow*, Appl. Math. Model., vol. 40, no. 4, pp. 2999–3010, 2016. DOI: 10.1016/j.apm.2015.09.088.

## Lecture Series

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|--------------------|----------------------------------------------------------------------------------------------------------------------|
| June 16–20, 2025   | Étale cohomology and étale homotopy, Goethe-Universität Frankfurt<br><i>Étale homotopy theory and exodromy</i>       |
| June 24–28, 2024   | Masterclass on Exit Paths and Stratified Homotopy Types, Københavns Universitet<br><i>Exodromy beyond conicality</i> |
| August 16–20, 2023 | Symplectic Paradise Summer School, Paradise Valley, Montana<br><i>From stratifications to singular Lagrangians</i>   |

## Selected Talks

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| October 9, 2025    | Young Homotopy Theorist Meeting, ICMS, Edinburgh<br><i>The condensed homotopy type of a scheme</i>                              |
| June 10, 2025      | Modern Developments in Geometry and Higher Structures, Brin Center, University of Maryland<br><i>The proétale homotopy type</i> |
| January 12, 2025   | JMM Satellite Session on Homotopy Theory, University of Washington<br><i>Applications of exodromy in topology</i>               |
| November 12, 2024  | Algebraic Geometry Seminar, Johns Hopkins<br><i>Reconstructing schemes from their étale topoi</i>                               |
| November 4, 2024   | Colloquium, University of Southern California<br><i>Exit-path categories in topology and algebraic geometry</i>                 |
| October 28, 2024   | RTG Seminar on Arithmetic Geometry, UC Berkeley<br><i>Weight filtrations on the Betti cohomology of complex varieties</i>       |
| September 2024     | Anabelian and Motivic Homotopy Theory, Universität Heidelberg<br><i>Reconstructing schemes from their étale topoi</i>           |
| September 17, 2024 | Seminar on Commutative Algebra and Algebraic Geometry, UC Berkeley<br><i>Reconstructing schemes from their étale topoi</i>      |
| September 2024     | Toposes in Mondovì, Grothendieck Institute<br><i>Reconstructing schemes from their étale topoi</i>                              |

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| August 27, 2024   | <b>Enumerative Geometry Beyond Spaces</b> , Banff International Research Station<br><i>Interactions between homotopy theory and algebraic geometry</i>                                   |
| May 9, 2024       | <b>Szygies and Mirror Symmetry Virtual Seminar</b><br><i>An introduction to “Exodromy beyond conicality”</i>                                                                             |
| April 17, 2024    | <b>Algebraic Geometry Seminar</b> , UC Davis<br><i>Exodromy in topology and applications</i>                                                                                             |
| March 17, 2024    | <b>Lloyd Roeling Conference</b> , University of Louisiana at Lafayette<br><i>Exodromy in topology and applications</i>                                                                   |
| December 12, 2023 | <b>Category Theory Seminar</b> , Johns Hopkins<br><i>How to invent pyknotic/condensed mathematics</i>                                                                                    |
| December 11, 2023 | <b>Topology Seminar</b> , Johns Hopkins<br><i>Exodromy beyond conicality</i>                                                                                                             |
| November 6, 2023  | <b>Algebra Seminar</b> , University of Southern California<br><i>On Voevodsky’s reconstruction theorem</i>                                                                               |
| October 20, 2023  | <b>Algebra &amp; Number Theory Seminar</b> , University of California Santa Cruz<br><i>Weight filtrations from the perspective of homotopy theory</i>                                    |
| June 20, 2023     | <b>Arbeitstagung 2023 on Condensed Mathematics</b> , Max-Planck-Institut für Mathematik, Bonn<br><i>Condensed math &amp; exodromy</i>                                                    |
| May 5, 2023       | <b>Algebraic Topology Seminar</b> , Columbia<br><i>Spectral weight filtrations</i>                                                                                                       |
| April 24, 2023    | <b>Topology Seminar</b> , MIT<br><i>Spectral weight filtrations</i>                                                                                                                      |
| March 20, 2023    | <b>Mathematics Seminar</b> , Montana State University<br><i>Reconstruction and étale homotopy theory</i>                                                                                 |
| February 16, 2023 | <b>Geometry-Topology Seminar</b> , University of Pennsylvania<br><i>Reconstruction and étale homotopy theory</i>                                                                         |
| January 13, 2023  | <b>Spring Opportunities Workshop</b> , Institute for Advanced Study<br><i>Reconstruction in algebraic geometry</i>                                                                       |
| October 18, 2022  | <b>Topology Seminar</b> , University of California San Diego<br><i>New perspectives on the étale homotopy type</i>                                                                       |
| July 28, 2022     | <b>Oberseminar: Higher Topoi and Exodromy</b> , Universität Regensburg<br><i>Microlocal sheaves</i>                                                                                      |
| July 7, 2022      | <b>Derived Categories, Arithmetic, and Reconstruction in Algebraic Geometry</b> , Banff International Research Station<br><i>Galois-theoretic reconstruction of schemes and exodromy</i> |

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| May 13, 2022      | <b>Algebraic Topology Seminar, UCLA</b><br><i>New perspectives on the reconstruction of schemes and étale homotopy types</i>                                |
| May 6, 2022       | <b>Homotopy Theory Seminar, University of Pennsylvania</b><br><i>Revisiting classical splitting results</i>                                                 |
| March 28, 2022    | <b>MUSA Math Monday, UC Berkeley</b><br><i>The cobordism hypothesis in low dimensions</i>                                                                   |
| February 2, 2022  | <b>Oberseminar: Higher Topoi and Exodromy, Universität Regensburg</b><br><i>Exit-path categories in algebra and geometry</i>                                |
| January 24, 2022  | <b>Algebra &amp; Geometry Seminar, Caltech</b><br><i>Stratified étale homotopy theory</i>                                                                   |
| January 11, 2022  | <b>Research Seminar on Algebraic Topology, Universität Hamburg</b><br><i>Exodromy for <math>\ell</math>-adic sheaves (or how I got into condensed math)</i> |
| November 11, 2021 | <b>Algebraic &amp; Geometric Topology Seminar, University of Nevada, Reno</b><br><i>Exit-path categories in geometry and topology</i>                       |
| October 30, 2021  | <b>Talbot 2020One: Ambidexterity in Chromatic Homotopy Theory</b><br><i>Lecture 16: Character isomorphisms via tempered cohomology</i>                      |
| May 3, 2021       | <b>Topology Seminar, MIT</b><br><i>On the homotopy theory of stratified spaces</i>                                                                          |
| February 11, 2021 | <b>Thursday Seminar: Redshift in algebraic K-theory, Harvard</b><br><i>Introduction to algebraic K-theory</i>                                               |
| December 14, 2020 | <b>Research Seminar on Algebraic Topology, Universität Hamburg</b><br><i>Revisiting classical splitting results</i>                                         |
| December 7, 2020  | <b>Algebraic Topology Seminar, University of Michigan</b><br><i>Stratified étale homotopy theory</i>                                                        |
| November 3, 2020  | <b>UChicago-Northwestern Topology Seminar</b><br><i>Stratified étale homotopy theory</i>                                                                    |
| June 2, 2020      | <b>Oberseminar: Integral Homotopy Theory, Universität Regensburg</b><br><i>Borel Global Algebras</i>                                                        |
| May 20, 2020      | <b>Motives and What Not, Motivic Zoom Conference Series</b><br><i>Stratified étale homotopy theory</i>                                                      |
| May 5, 2020       | <b>SFB Lecture, Universität Regensburg</b><br><i>New perspectives on étale homotopy theory</i>                                                              |
| April 28, 2020    | <b>Working Group on Stratified Homotopy Theory, MSRI</b><br><i>The homotopy theory of stratified spaces</i>                                                 |
| April 24, 2020    | <b>Working Group on Stratified Homotopy Theory, MSRI</b><br><i>Intro to constructible sheaves &amp; exit paths</i>                                          |

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| March 10, 2020     | <b>Pyknotic/Condensed Seminar, MSRI</b><br><i>Pyknotic spaces</i>                                                                                  |
| March 4, 2020      | <b>Pyknotic/Condensed Seminar, MSRI</b><br><i>Pyknotic sets</i>                                                                                    |
| December 2, 2019   | <b>Topology Seminar, Johns Hopkins</b><br><i>On the homotopy theory of stratified spaces</i>                                                       |
| October 13, 2019   | <b>AMS Fall Eastern Sectional Meeting, Binghamton, NY</b><br><i>On the homotopy theory of stratified spaces</i>                                    |
| September 10, 2019 | <b>Topology Seminar, Notre Dame</b><br><i>Stratified spaces, constructible sheaves, &amp; exit-paths</i>                                           |
| August 8, 2019     | <b>Workshop on Homotopy Theory, Mathematisches Forschungsinstitut Oberwolfach</b><br><i>On the homotopy theory of stratified spaces</i>            |
| October 18, 2018   | <b>Homotopy Harnessing Higher Structures Programme, Isaac Newton Institute</b><br><i>Constructible étale &amp; analytic sheaves and exit paths</i> |
| December 7, 2017   | <b>Homotopy Theory Seminar, The Ohio State University</b><br><i>Monodromy &amp; Stratified Homotopy Theory</i>                                     |

### Teaching Experience

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|-------------|------------------------------------------------------------------------------------------------------------|
| Spring 2026 | <b>Math 250B, Topics in Algebraic Geometry: Mirror Symmetry for Toric Varieties</b> (graduate course), USC |
| Fall 2025   | <b>Math 226, Calculus III</b> , USC                                                                        |
| Spring 2025 | <b>Math 250B, Commutative Algebra</b> (graduate course), UC Berkeley                                       |
| Fall 2024   | <b>Math 256A, Algebraic Geometry</b> (graduate course), UC Berkeley                                        |
| Spring 2024 | <b>Math 250B, Commutative Algebra</b> (graduate course), UC Berkeley                                       |
| Fall 2023   | <b>Math 256A, Algebraic Geometry</b> (graduate course), UC Berkeley                                        |
| Fall 2019   | <b>18.821, Project Laboratory in Mathematics, Teaching Assistant</b> , MIT                                 |

### Mentoring Experience

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| Spring 2025 | <b>Undergraduate reading course on topos theory &amp; categorical logic</b> , UC Berkeley                                                                 |
| Fall 2024   | <b>Undergraduate reading course on homotopy theory &amp; higher category theory</b> , UC Berkeley                                                         |
| Spring 2024 | <b>Undergraduate reading course on homotopy theory &amp; higher category theory</b> , UC Berkeley                                                         |
| Spring 2022 | <b>Undergraduate reading course on homotopy theory &amp; higher category theory</b> , UC Berkeley                                                         |
| 2020–2021   | <b>Mentor</b> , MIT Math Department's MIT Grad-Undergrad Math Mentoring Initiative (GUMMI)<br><i>Mentored two undergrads applying to math grad school</i> |

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| 2018–2019   | <b>Reading Course Mentor</b> , MIT Department of Mathematics<br><i>Mentored an undergraduate reading course on differential forms and equivariant de Rham theory with Victor Guillemin</i> |
| Winter 2018 | <b>Directed Reading Program Mentor</b> , MIT Department of Mathematics<br><i>Mentored an undergraduate in a category theory and topos theory reading course</i>                            |
| Summer 2017 | <b><math>\sqrt{\text{Math}}</math>roots Academic Mentor</b><br><i>Mentored high school students through the MIT Math Department's <math>\sqrt{\text{Math}}</math>roots program</i>         |
| Winter 2017 | <b>Directed Reading Program Mentor</b> , MIT Department of Mathematics<br><i>Mentored two undergraduates in reading Emily Riehl's text Category theory in Context</i>                      |

### Service & Organizational Activities

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| Fall 2024        | <b>Faculty co-sponsor</b> , MUSA 174, Student-taught course on category theory, UC Berkeley                         |
| 2021–2022        | <b>Co-organizer</b> , Nadler's Geometric Representation Theory Seminar                                              |
| Fall 2020        | <b>Co-organizer</b> , Harvard/MIT Thursday Seminar on Condensed Mathematics                                         |
| Fall 2020        | <b>Co-organizer</b> , MIT Juvitop Seminar on the Cobordism Hypothesis                                               |
| 2020–2021        | <b>Co-organizer</b> , MIT Grad-Undergrad Math Mentoring Initiative (GUMMI)                                          |
| Summer 2019      | <b>Co-organizer</b> , MIT Reading Group on <i>An Inclusive Academy</i>                                              |
| 2019–2021        | <b>School of Science Graduate Student Council</b> , MIT                                                             |
| Fall 2019        | <b>Co-organizer</b> , MIT Juvitop Seminar on Differential Cohomology                                                |
| Spring 2019      | <b>Co-organizer</b> , Miniature Seminar on Factorization Homology                                                   |
| 2018–2021        | <b>Coordinator</b> , MIT PRIMES Circle                                                                              |
| Fall 2018        | <b>Co-organizer</b> , MIT Juvitop Seminar on <i>Ambidexterity in <math>K(n)</math>-local Stable Homotopy Theory</i> |
| Fall 2017        | <b>Organizer</b> , MIT Topology Seminar                                                                             |
| Spring 2016–2021 | <b>Diversity and Community Building Committee</b> , MIT Department of Mathematics                                   |